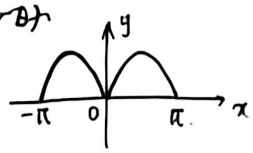
Exe. Chapter 2.

4. Conrider the 2π-periodic function defined on $[0,\pi]$ by $f(\theta)=\theta(\pi-\theta)$ (even. (1.10)).

(b) compute the fourier series of f and show that $f(\theta)=\frac{8}{\pi}\sum_{k \text{ even}\geq 1} \frac{\sin k\theta}{k^3}\cdot\frac{\cos k\theta}{k^2}$

Pf) $\hat{f}(n)=\frac{1}{2\pi}\int_{-\pi}^{\pi} f(\theta)e^{-in\theta}d\theta$.



$\frac{1}{2\pi}\left(\int_0^{\pi} f(\theta)\,e^{-in\theta}\,d\theta+\int_{-\pi}^{0} f(\theta)\,e^{-in\theta}\,d\theta\right)$

$=\frac{1}{2\pi}\int_0^{\pi}\theta(\pi-\theta)\left(e^{-in\theta}+e^{in\theta}\right)d\theta$.

$=\frac{1}{\pi}\int^{\pi}\theta\cdot(\pi-\theta)\cos n\theta\,d\theta=\frac{1}{\pi n}\int_0^{\pi}\theta(\pi-\theta)\,d(\sin n\theta)$

$=-\frac{1}{\pi n}\int_0^{\pi}(\pi-2\theta)\sin n\theta\,d\theta$.

$=\frac{1}{n^2\pi}\int_0^{\pi}(\pi-2\theta)\,d(\cos n\theta)$

$=\frac{1}{n^2\pi}(\pi-2\theta)\cos n\theta\Big|_0^{\pi}+\frac{2}{n^2\pi}\int_0^{\pi}\cos n\theta\,d\theta$ ($\frac{-1}{n}\sin n\theta\Big|_0^{\pi}$)

$=0$ $(2\nmid n)$ or $\frac{-2}{n^2\pi}$ $(2\mid n)\Rightarrow\sum_{|n|\geq 1}|\hat{f}(n)|<\infty$.

$\Rightarrow f(\theta)=\sum_{N=-\infty}^{\infty}\hat{f}(n)e^{in\theta}=-\sum_{N=-\infty}^{\infty}\frac{2}{N^2}\cos N\theta=-4\sum_{N=1}^{\infty}\frac{\cos N\theta}{N^2}$ (even)

then $-4\sum_{N=1}^{\infty}\frac{\cos N\theta}{N^2}=-\frac{8}{\pi}\sum_{N=1}^{\infty}\frac{\sin N\theta}{N^3}=-4\sum_{N=1}^{\infty}\frac{1}{N^2}\left(\cos N\theta+\frac{2}{\pi}\frac{\sin N\theta}{N}\right)$ □

(even) (even).

5. Consider the function: $f(\theta)=\begin{cases}0 & \text{if } |\theta|>\delta\\ 1-\frac{|\theta|}{\delta} & \text{if } |\theta|\leq\delta\end{cases}$

Show that $f(\theta)=\frac{\delta}{2\pi}+2\sum_{n=1}^{\infty}\frac{1-\cos n\delta}{n^2\pi\delta}\cos n\theta$.

$\hat{f}(0)=\frac{1}{2\pi}\int_0^{\delta}\left(1-\frac{\theta}{\delta}\right)d\theta+\frac{1}{2\pi}\int_{-\delta}^{0}\left(1+\frac{\theta}{\delta}\right)d\theta=\frac{\delta}{2\pi}$

Pf: $\hat{f}(n)=\frac{1}{2\pi}\int_{-\pi}^{\pi} f(\theta)\,e^{-in\theta}d\theta=\frac{1}{2\pi}\int_{|\theta|\leq\delta}\left(1-\frac{|\theta|}{\delta}\right)e^{-in\theta}\,d\theta$.

$\frac{1}{2\pi}\int_0^{\delta}\left(1-\frac{\theta}{\delta}\right)\left(e^{-in\theta}+e^{in\theta}\right)d\theta$

$=\frac{1}{\pi}\int_0^{\delta}\left(1-\frac{\theta}{\delta}\right)\cos n\theta\,d\theta$

$=\frac{1}{n\pi}\left(1-\frac{\theta}{\delta}\right)\sin n\theta\Big|_0^{\delta}-\frac{1}{n\pi}\int_0^{\delta}\cdot\left(\frac{1}{-\delta}\right)\sin n\theta\,d\theta$

$=\frac{1}{n^2\pi\delta}(\cos n\delta+1)$ $(n\neq 0)$

then $f(\theta)=\sum_{N=-\infty}^{\infty}\hat{f}(n)e^{in\theta}=2\sum_{N=1}^{\infty}\frac{(1-\cos n\delta)\cos n\theta}{n^2\pi\delta}+\frac{\delta}{2\pi}$ □ $\left(\sum|\hat{f}(n)|<\infty\right)$

6. f defined on $[-\pi,\pi]$. by $f(\theta)=|\theta|$

(b). caculate the fourier series of f. show that $\hat{f}(n)=\begin{cases}\frac{\pi}{2} & n=0\\ \frac{-1+(-1)^n}{n^2\pi} & n\neq 0\end{cases}$

(d). Taking $\theta=0$, prove that $\sum_{n\,\text{odd}\geq 1}\frac{1}{n^2}=\frac{\pi^2}{8}$ and $\sum_{n\geq 1}^{\infty}\frac{1}{n^2}=\frac{\pi^2}{6}$

pf: (b) $\hat{f}(0)=\frac{1}{2\pi}\int_{-\pi}^{\pi}|\theta|\,d\theta=\frac{\pi}{2}$

$$\hat{f}(n)=\frac{1}{2\pi}\int_{-\pi}^{\pi}|\theta|e^{-in\theta}d\theta=\frac{1}{2\pi}\int_0^{\pi}\theta\left(e^{-in\theta}+e^{in\theta}\right)d\theta$$

$$=\frac{1}{\pi}\int_0^{\pi}\theta\cos n\theta\,d\theta=\frac{1}{n\pi}\theta\sin n\theta\Big|_0^{\pi}-\frac{1}{n\pi}\int_0^{\pi}\sin n\theta\,d\theta$$

$$=\frac{1}{n^2\pi}\left[(-1)+(-1)^n\right]\quad(n\neq 0)$$

$\sum|\hat{f}(n)|<\infty$

Then: $f(\theta)=\sum_{N=-\infty}^{\infty}\hat{f}(N)e^{iN\theta}=\sum_{\substack{N=1\\(\text{odd})}}^{\infty}\frac{-4}{N^2\pi}\cos n\theta+\frac{\pi}{2}$

(d) $f(0)=0=-\sum_{\substack{N=1\\(\text{odd})}}^{\infty}\frac{4}{N^2\pi}+\frac{\pi}{2}\Rightarrow\sum_{\substack{N=1\\(\text{odd})}}^{\infty}\frac{1}{N^2}=\frac{\pi^2}{8}$.

also: $\sum_{n=1}^{\infty}\frac{1}{n^2}=\sum_{n(\text{odd})\geq 1}\frac{1}{n^2}+\sum_{n(\text{even})\geq 1}\frac{1}{n^2}=\frac{\pi^2}{8}+\frac{1}{4}\sum_{n=1}^{\infty}\frac{1}{n^2}\Rightarrow\sum_{n=1}^{\infty}\frac{1}{n^2}=\frac{\pi^2}{6}$ □.

8. Verify that $\frac{1}{2i}\sum_{n\neq 0}\frac{e^{inx}}{n}$ is the fourier series of the 2π-periodic saw tooth function defined by $f(0)=0$,

$$f(x)=\begin{cases}-\frac{\pi}{2}-\frac{x}{2} & x\in(-\frac{\pi}{2},0)\\ \frac{\pi}{2}-\frac{x}{2} & x\in(0,\frac{\pi}{2}).\end{cases}$$

$\hat{f}(0)=0$. $\hat{f}(n)=\frac{1}{2\pi}\int_{-\pi}^{\pi}f(\theta)e^{-in\theta}d\theta=\frac{1}{2\pi}\left(\int_0^{\pi}(\frac{\pi}{2}-\frac{\theta}{2})e^{-in\theta}d\theta+\int_{-\pi}^{0}(-\frac{\pi}{2}-\frac{\theta}{2})e^{-in\theta}d\theta\right)$

$$=\frac{1}{2\pi}\int_0^{\pi}(\tfrac{\pi}{2}-\tfrac{\theta}{2})(e^{-in\theta}-e^{in\theta})\,d\theta.$$

$$=\frac{-1}{2\pi i}\int_0^{\pi}(\pi-\theta)\sin n\theta\,d\theta.$$

$$=\frac{1}{2\pi ni}\left[(\pi-\theta)\cos n\theta\Big|_0^{\pi}+\int_0^{\pi}\cos n\theta\,d\theta\right]$$

$$=-\frac{1}{2ni}$$

$\Rightarrow f(x)\sim\sum_{n\neq 0}\hat{f}(n)e^{inx}=\frac{1}{2i}\sum_{n\neq 0}\frac{e^{inx}}{n}$.

$\forall x\in\mathbb{R}$. $\frac{1}{2i}\sum_{n\neq 0}\frac{e^{inx}}{n}=\sum_{n=1}^{\infty}\frac{\sin nx}{n}$.

$S_N-S_M=\sum_{n=M}^{N}\frac{1}{n}\sin nx=\frac{\sin Nx}{N}-\frac{\sin(M-1)x}{M}-\sum_{n=M}^{N-1}\left(\frac{1}{n+1}-\frac{1}{n}\right)\sin nx<\varepsilon$. when M,N is large enough

So we have $\sum_{n=1}^{\infty}\frac{\sin nx}{n}$ converges for every x.

9. $f(x) = \chi_{[a,b]}(x) = \begin{cases} 1 & x \in [a,b] \\ 0 & \text{otherwise} \end{cases}$ $\quad [a,b] \subsetneq [-\pi,\pi]$.

prove:

(a) $f(x) \sim \frac{b-a}{2\pi} + \sum_{n\neq 0} \frac{e^{-ina} - e^{-inb}}{2\pi i n} e^{inx}$.

$\hat{f}(0) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x) e^{-i0x}\,dx = \frac{1}{2\pi}\int_a^b 1\,dx = \frac{b-a}{2\pi}$.

$\hat{f}(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x) e^{-inx}\,dx = \frac{1}{2\pi}\int_a^b e^{-inx}\,dx = \frac{1}{2\pi i n}(e^{-ina} - e^{-inb})$. □.

(b) if $a \neq -\pi$ / $b \neq \pi$, then the fourier series doesn't converge absolutely for any x.

$S_n = \sum_{n\neq 0} \frac{e^{-ina} - e^{-inb}}{2\pi i n} e^{inx} = \sum_{n=1}^{\infty} \frac{\sin nb - \sin na}{\pi n}\cos nx + \sum_{n=1}^{\infty} \frac{\cos na - \cos nb}{\pi n}\sin nx$

$\sum_{n\neq 0} \frac{e^{in(x-a)}}{2\pi i n} (= \sum_{n\neq 0} \frac{e^{in(x-b)}}{2\pi i n}$.

$= \sum_{n=1}^{\infty} \frac{1}{2\pi i n}(e^{in(x-a)} - e^{-in(x-a)}) - \sum_{n=1}^{\infty} \frac{1}{2\pi i n}(e^{in(x-b)} - e^{-in(x-b)})$

$= \sum_{n=1}^{\infty} \frac{1}{\pi n}(\sin n(x-a) - \sin n(x-b)) = \sum_{n=1}^{\infty} \frac{2\sin n\theta_0 \cos n(x - \frac{a+b}{2})}{n}$.

$\theta_0 = \frac{b-a}{2} \neq \pi$,

$\Rightarrow \sum_{n=1}^{\infty} \frac{2|\sin n\theta_0|\,|\cos n(x - \frac{a+b}{2})|}{n}$ that's wrong!

真正的 absolute convergence: $\sum_{n\neq 0} |c_n e^{inx}| = 2\sum_{n=1}^{\infty} |c_n| \geq \sum_{n=1}^{\infty} |c_n| = \sum_{n=1}^{\infty} \left|\frac{e^{-ina}(1 - e^{-ind})}{2\pi i n}\right|$

$= \sum_{n=1}^{\infty} \frac{|e^{in\frac{d}{2}} - e^{-in\frac{d}{2}}|}{2\pi |n|}$

$= \sum_{n=1}^{\infty} \frac{\sin|n\frac{d}{2}|}{2\pi|n|}$ $\quad d = \frac{b-a}{2} \neq \pi$

There exists many n s.t $|\sin n\frac{d}{2}| \geq c > 0$. (with Weil Thm: $S = \{n : |\sin nd| \geq c > \frac{1}{2}\}$ $|S|$ have the positive p.

Then $\sum_{n=1}^{\infty} \frac{\sin|\frac{nd}{2}|}{2\pi|n|} \to \infty$ □.

(c). pf: $\sum_{n\neq 0}$ * the fourier series converges at every point of x And consider $\begin{cases} a = -\pi \\ b = \pi \end{cases}$.

pf: 1) $S_n(f)(x) = \sum_{n=1}^{\infty} \frac{\sin n(x-a)}{\pi n} - \sum_{n=1}^{\infty} \frac{\sin n(x-b)}{\pi n} + \frac{b-a}{2\pi}$ with the Dirchlet's test for $\sum a_n b_n$

It's obvious.

2). $\begin{cases} a = -\pi \\ b = \pi \end{cases}$ then $S_n(f)(x) = \frac{b-a}{2\pi}$ □.

10. Suppose f is 2π-periodic and $f \in C^k[-\pi,\pi]$. prove $\hat{f}(n) = O\left(\frac{1}{|n|^k}\right)$ as $|n| \to \infty$

pf: $\hat{f}(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\, e^{-inx}\, dx = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\, d\left(\frac{e^{-inx}}{-in}\right) = -\frac{1}{2\pi in}\int_{-\pi}^{\pi} f'(x)\, e^{-inx}\, dx$.

$= \cdots = (-1)^k \frac{1}{2\pi (in)^k}\int_{-\pi}^{\pi} f^{(k)}(x)\, e^{-inx}\, dx$.

$= O\left(\frac{1}{|n|^k}\right)$ as $|n| \to \infty$ □

11. Suppose that $\{f_k\}_{k=1}^{\infty}$ is a sequence of R-integrable fs. on $[0,1]$: that $\int_0^1 |f_k(x) - f(x)|\, dx \to 0$ $(k \to \infty)$

Show that $\hat{f}_k(n) \to \hat{f}(n)$ uniformly. in n. as $k \to \infty$

pf: $\hat{f}_k(n) = \int_0^1 f_k(x)\, e^{-2\pi i n x}\, dx$. $\hat{f}(n) = \int_0^1 f(x)\, e^{-2\pi i n x}\, dx$.

$\Rightarrow |\hat{f}_k(n) - \hat{f}(n)| \le \int_0^1 |f_k(x) - f(x)|\, dx$. let $k \to \infty$ then $\lim_{k\to\infty} \hat{f}_k(n) = \hat{f}(n)$ (uniformly). □

13. Show that Convergent → Cesàro summable → Abel summable.

1). $S_n = \sum_{k=0}^{n} c_k$. $\sigma_n = \frac{1}{n}\sum_{k=0}^{n} S_k$. $A(r) = \sum_{k=0}^{\infty} c_k r^k$

With Cauchy Prop. it's obvious that : if $S_n \to S$. then $\sigma_n \to S$.

2) if $S_n \to S$. then $\sum_{k=0}^{\infty} c_k r^k = (1-r)\sum_{k=1}^{N} S_k r^k + S_N r^{N+1}$ (Abel Summation).

So $A(r)$ converges $A(r) = (1-r)\sum_{k=1}^{\infty} S_k r^k$.

if $S_k \to S = 0$ $(k \to \infty)$ then $\forall \varepsilon > 0$, let N big. r small. s.t
(it's enough)

$$A(r) = (1-r)\sum_{k=1}^{N} S_k r^k + (1-r)\sum_{k=N+1}^{\infty} S_k r^k$$

$$< \varepsilon + \varepsilon (1-r)\cdot \frac{1}{1-r} = 2\varepsilon \quad \square$$

3). if $\sigma_n \to S$. then $\sum_{k=1}^{N} c_k r^k = (1-r)\sum_{k=1}^{N} S_k r^k + S_N r^{N+1}$

$$= (1-r)^2 \sum_{k=1}^{N} k\sigma_k r^k + N\sigma_N r^{N+1} + S_N r^{N+1}$$

let $N \to \infty$. then $A(r) = (1-r)^2 \sum_{n=1}^{\infty} n\sigma_n r^n$

Assume $S = 0$ $\forall \varepsilon > 0$. $\exists N$, $\forall n > N$ 有 $|\sigma_n| < \varepsilon$. 取 $\delta > 0$. s.t. $|r-1| < \delta$ 时 $(1-r^2)^2 \sum_{n=1}^{N} n\sigma_n r^n < \varepsilon$

$$A(r) = (1-r)^2 \sum_{n=1}^{N} n\sigma_n r^n + (1-r)^2 \sum_{n=N+1}^{\infty} n\sigma_n r^n$$

$$< \varepsilon + \varepsilon (1-r)^2 \sum_{n=N+1}^{\infty} n r^n < \varepsilon + \varepsilon r < 2\varepsilon. \quad \square.$$

16. Let $f \in C([a,b])$ Then for any $\varepsilon>0$ there exists a polynomial P such that $\sup_{x\in[a,b]} |f(x)-p(x)| < \varepsilon$.

pf: Lemma: If $f \in C(\mathbb{R}/2\pi\mathbb{R})$, then it can be approximated by trigonometric poly.

pf: for $f \in C(\mathbb{R}/2\pi\mathbb{R})$. so. $\lim_{n\to\infty} \sigma_n(f)(x) = f(x)$ uniformly in x

$$\sigma_n(f)(x) = (f * F_N)(x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} F_N(y) f(x-y)\,dy. = \frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{1}{N}\sum_{n=0}^{N-1} D_n(x-y) f(y)\,dy$$

$$= \sum_{k=-n}^{n} \left(1-\frac{|k|}{n}\right) \hat{f}(k) e^{ikx}$$

since: $D_k(x) = \frac{\sin(k+\frac{1}{2})x}{\sin\frac{1}{2}x}$ are trig-poly. then $\sigma_n(f)(x)$ is trig-poly. ☒

back to the proof: $\forall \varepsilon>0.\ \exists N>0:$ $|f(x) - \sigma_N(f)(x)| < \frac{\varepsilon}{4}$.

$\sigma_N(f)(x) := T(x) \sim Q(x)$ 对 e^{ikx} can be approximated by poly ...

Then $|f(x)-Q(x)| < |f(x)-T(x)| + |T(x)-Q(x)| < \varepsilon$ ☒.

17. a). Prove that: If f has a jump discountinuity at θ.

then $\lim_{r\to 1^-} A_r(f)(\theta) = \frac{1}{2}(f(\theta^+) + f(\theta^-))$, with $0 \le r < 1$

pf:

$$\lim_{r\to1} A_r(f)(\theta) = \lim_{r\to1} (f * P_r)(\theta) = \lim_{r\to1} \frac{1}{2\pi}\int_{-\pi}^{\pi} f(\varphi)\left(\sum_{n=-\infty}^{\infty} r^{|n|} e^{-in(\varphi-\theta)}\right) d\varphi.$$

$$= \lim_{r\to1} \frac{1}{2\pi}\int_{-\pi}^{0} f(\varphi) P_r(\varphi-\theta)\,d\varphi + \lim_{r\to1}\frac{1}{2\pi}\int_{0}^{\pi} f(\varphi) P_r(\varphi-\theta)\,d\varphi$$

$$\lim_{r\to1} \left|\frac{1}{2\pi}\int_{-\pi}^{0} P_r(\varphi)(f(\theta-\varphi)-f(\theta^-))\,d\varphi\right| \doteq \left|\frac{1}{2\pi}\int_{-\pi}^{0} f(\theta-\varphi) P_r(\varphi)\,d\varphi - f(\theta^-)\frac{1}{2}\right|$$

$$\le \frac{1}{2\pi}\int_{-\pi}^{0} |P_r(\varphi)|\,|f(\theta-\varphi)-f(\theta^-)|\,d\varphi.$$

$$= \frac{1}{2\pi}\left(\int_{-\pi\le\varphi\le-\delta} + \int_{-\delta\le\varphi\le0}\right)\left(|P_r(\varphi)|\cdot|f(\theta-\varphi)-f(\theta^-)|\right) d\varphi$$

$\forall \varepsilon>0.\ \exists r>0.$ s.t $\int_{|\varphi|\ge\delta} |P_r(\varphi)| < \varepsilon$.

$\exists \delta>0.$ s.t. $|f(\theta-\varphi)-f(\theta^-)| < \varepsilon$.

$\Rightarrow \left|\frac{1}{2\pi}\int_{-\pi}^{0} f(\theta-\varphi) P_r(\varphi)\,d\varphi - \frac{1}{2}f(\theta^-)\right| < B\varepsilon$. B is relevant to θ & r & ε.

Thus. $\lim_{r\to1} A_r(f)(\theta) = \lim_{r\to1}(f*P_r)(\theta) = \frac{1}{2}(f(\theta^-)+f(\theta^+))$. ☒